

# JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 218 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                     | Frequency | Category        |
|---------------|--------------------------------------------------|-----------|-----------------|
| Very High     | Permutations And Combinations - Core Application | 218 PYQs  | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Permutations And Combinations - Core Application (Very High Yield — 218 PYQs)

Core Principle: Key problem pattern in Permutations And Combinations.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag i..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let  $A=(a,b,c):a,b,c$  are non-negative integers and  $a+b+2c=22$  . Then  $n(A)$  is equal to :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The number of elements in the set  $S=(r,k):k \in \mathbb{N}$  and  ${}^{36}C_{(r+1)}=\frac{6({}^{35}C_{(r)})(k^2-3)}$  is :..." [Ans: B]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)**

The number of elements in the set  $S = \{(r, k) : k \in \mathbb{N} \text{ and } {}^{36}C_{r+1} = \frac{{}^{35}C_r}{k^2 - 3}\}$  is :

- (A) 2
- (B) 4 **[Correct]**
- (C) 8
- (D) 16

**Correct Answer: B**

**Solution Analysis:** We need to find the number of ordered pairs  $(r, k)$  satisfying  ${}^{36}C_{r+1} = \frac{{}^{35}C_r}{k^2 - 3}$ , where  $k \in \mathbb{N}$ . Let us simplify the relation using the standard NCERT combination formula. We know that  ${}^nC_{r+1} = \frac{n!}{(r+1)!(n-r-1)!}$  and  ${}^{n-1}C_r = \frac{(n-1)!}{r!(n-r-1)!}$ . So,  $\frac{{}^{36}C_{r+1}}{{}^{35}C_r} = \frac{(36! / (r+1)!(35-r)!)}{(35! / r!(35-r)!)} = \frac{36!}{35!} \cdot \frac{r!}{(r+1)!} = 36 \cdot \frac{1}{r+1} = \frac{36}{r+1}$ . Hence,  ${}^{36}C_{r+1} = {}^{35}C_r \cdot \frac{36}{r+1}$ . Given,  ${}^{36}C_{r+1} = \frac{{}^{35}C_r}{k^2 - 3}$ . Comparing both expr